Calculus Complete Notes

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CALCULUS - COMPLETE NOTES

1. LIMITS
lim(x->a) f(x) = L
Key Laws:
lim [f+g] = lim f + lim g
lim [f g] = lim f lim g

Standard:
lim(x->0) sin(x)/x = 1
lim(x->0) (1-cos x)/x = 0
lim(x->0) (e^x - 1)/x = 1

2. DERIVATIVES
f'(x) = lim(h->0) [f(x+h)-f(x)]/h

Power: d/dx(x^n) = n x^(n-1)
Product: (uv)' = u'v + uv'
Chain: dy/dx = dy/du du/dx

d/dx(sin x) = cos x
d/dx(cos x) = -sin x
d/dx(e^x) = e^x
d/dx(ln x) = 1/x

3. INTEGRALS

4. FTC

5. SERIES
e^x = 1 + x + x /2! + x /3! + ...